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SMARAG: a trustworthy AI system for carbon reporting using self-decisive multi-agentic retrieval-augmented generation

Xinlai Liu [a], Praneet Pabolu [b], Yishuo Jiang [a], Timothy Fraser [a], Runying Chen [c], H. Oliver Gao [a,d,*]

[a] Systems Engineering Program, Cornell University, Ithaca, NY 14853, the United States of America
[b] Department of Computer Science, Cornell University, Ithaca, NY 14853, the United States of America
[c] Thomas M. Siebel Center for Computer Science, University of Illinois Urbana-Champaign, 1201 West University Avenue, Urbana, IL 61801, the United States of America
[d] School of Civil and Environmental Engineering, Cornell University, Ithaca, NY 14853, the United States of America

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ABSTRACT

Carbon reporting is critical for assessing climate commitments, yet current methods are time-intensive and vulnerable to inconsistencies. While Large Language Models (LLMs) offer revolutionary potential in carbon reporting, LLMs struggle with hallucination issues, factual delay, and data security. This study proposes SMARAG, a Trustworthy Artificial Intelligence (TAI)-powered carbon reporting system that combines TAI design principles and Self-decisive Multi-Agentic Retrieval-Augmented Generation (RAG) to provide accurate, efficient, and trustworthy carbon reporting. Grounded in the EU Ethics Guidelines for Trustworthy AI, SMARAG replicates real-world carbon reporting workflows through a team of role-specialized autonomous agents, such as sustainability director, project manager, data analyst, report writer. The agents draw from a locally hosted knowledge repository that integrates public carbon standards, regulatory frameworks, and private organizational data, minimizing the risk of unauthorized data exposure to external LLMs. Moreover, SMARAG employs both a self-decisive validation loop and human-in-the-loop architecture to ensure outputs meet regulatory standards before finalization. Evaluated on real-life carbon data and standards, SMARAG outperforms multi-agentic RAG, single-agent (vanilla) RAG, and standalone LLMs, including GPT-4o, DeepSeek R1, Claude-3.5, Gemini-1.5, and Mistral-Instruct-v0.3, in terms of report quality, usefulness, faithfulness, and relevancy. It not only empowers resource-constrained organizations such as small and medium-sized enterprises to participate in climate governance, but also offers valuable insights for policymakers and practitioners committed to broader trustworthy sustainability disclosures and knowledge-intensive regulatory domains.

1. Introduction

Climate change is one of the most urgent global challenges of the 21st century, requiring collective action across communities, industries, and nations (Hoegh-Guldberg et al., 2019; Jia and Quaas, 2023). At the core of this action is carbon reporting, which is the process of measuring, documenting, and disclosing carbon emissions produced by economic units and carbon entities (Bellassen et al., 2015; Knox-Hayes and Levy, 2011). Carbon reporting gained momentum in the late 1990 s alongside voluntary climate disclosure initiatives and the Kyoto Protocol (Aiuto, Huckins and Momblanco, 2024). The 2015 Paris Agreement (United Nations Framework Convention on Climate Change, 2016) then elevated it to a near-universal obligation, committing nations to ambitious net-zero targets, including the United States (Kerry and Mccarthy, 2021), China (Yu et al., 2023), European Union (European Commission, 2019). Today, carbon reporting is no longer purely voluntary. Major economies have introduced mandatory disclosure regimes, such as European Union's Corporate Sustainability Reporting Directive (CSRD), United Kingdom's The Task Force on Climate-related Financial Disclosures (TCFD)-aligned climate reporting, and California Senate Bill 253, etc. These regulatory developments signal a decisive shift that carbon reporting has moved from a good practice to a legal requirement across

* Corresponding author at: Systems Engineering Program, Cornell University, Ithaca, NY 14853, the United States of America.
E-mail addresses: xl957@cornell.edu (X. Liu), pp542@cornell.edu (P. Pabolu), yj568@cornell.edu (Y. Jiang), tmf77@cornell.edu (T. Fraser), runying2@illinois.edu (R. Chen), hg55@cornell.edu (H.O. Gao).

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global markets, making its accuracy, consistency, and efficiency more consequential than ever.

Despite its critical role in fighting climate change, carbon reporting remains fraught with challenges. Firstly, carbon entities have to navigate diverse carbon standards and evolving policies that differ across jurisdictions, demanding adaptable reporting processes, boundaries, and compliance thresholds (Howard et al., 2015). For instance, multinational corporations must align and streamline carbon data collection, validation, and publication with varying location-specific rules while trying to reduce duplicating efforts (Gonenc and Poleska, 2022). As regulations evolve, the carbon entity is expected to swiftly update its carbon reporting. Secondly, carbon data sources are scattered across departments, systems, and regions, with varying standards complicating consolidation and comparability (Melville and Whisnant, 2014; Xu and MacAskill, 2024). Due to lack of interoperability (Schwanitz et al., 2022), it is difficult to produce consistent and comparable carbon reports. Thirdly, documentation for carbon reporting is a costly, time-consuming, and knowledge-intensive process, which relies on internal or external experts' experience. According to Ecohedge (ECHEDGE, 2023), the cost for carbon reporting varies from $20,000 to $200, 000 annually to have external consultants to take several weeks or months to measure, benchmark, and compute the carbon footprint depending on company scales and business complexity. These limitations not only hinder the quality and efficiency of carbon reporting but also increase the risk of greenwashing, undermining trust in climate finance (Sitarz et al., 2024).

Recent advancements in Artificial Intelligence (AI), particularly Large Language Models (LLMs), provide revolutionary potential to address the above carbon reporting challenges (Leal Filho et al., 2022). LLMs have demonstrated remarkable skills in natural language processing, content generation, and are increasingly applied in domains such as agriculture (Partha Pratim, 2023), manufacturing (Wang, Fan and Zheng, 2024), healthcare (Haltaufderheide and Ranisch, 2024), and finance (Lee, Stevens and Han, 2025), etc. However, the application of LLMs often falls short in well-defined tasks that require support of domain-specific knowledge like carbon reporting (Vaghefi et al., 2023; Wang et al., 2024), due to several endogenous and exogenous challenges:

(1) **Contextual Accuracy and Hallucinations**: LLMs can generate "hallucinations" or inaccurate information (Augenstein et al., 2024). A clear example is the chess game between ChatGPT-4o and DeepSeek-R1 organized by GothamChess (GothamChess, 2025) (A YouTuber with over six million of chess followers), where illegal and strategically unsound moves are repeatedly made, indicating their limitations in domain-specific knowledge/expertise. Accuracy is crucial for carbon reporting, making it important to ensure direct relevance of data retrieval and alignment of content generation with domain and industry specific knowledge and context.

(2) **The complex processes of carbon reporting**: Carbon reporting involves both internal and external stakeholders of a carbon entity (Zhou, 2020), such as sustainability officer, project manager, consultant manager, report writer, etc. Each of these stakeholders involved in the carbon reporting chain comes into play at different positions and may have different visions, knowledge base, and expectations for carbon reporting. It is hard to depend on the LLMs alone to understand all stakeholder expectations and workflows in carbon reporting, which are required to generate high-quality calculations and documentation.

(3) **Data privacy and security**: Carbon reporting involves sensitive information, such as fine-grained emissions data and production processes. It is promising to use AI to integrate such fine-grained data. However, there could be privacy and security concerns if carbon entity needs to directly provide all the related carbon data to LLMs (Preuss, Alshehri and You, 2024; Yao et al., 2024). Addressing this challenge requires a system that can handle sensitive organizational data locally, without exposing raw records to external AI providers.

These challenges reflect a broader imperative that scholars have articulated as Trustworthy AI (TAI). The principles of TAI emphasize that AI deployed in consequential domains must earn trust not through capability alone, but through demonstrable robustness, accountability, transparency, and respect for human agency and privacy (Floridi et al., 2018; Li et al., 2023; Tabassi, 2023). The Ethics Guidelines for Trustworthy AI published by the European Commission's High-Level Expert Group on Artificial Intelligence (HLEG, 2019) provide the most widely adopted international framework for operationalizing these principles, identifying seven concrete requirements that TAI systems should meet. We adopt the EU HLEG framework as the primary evaluative reference for SMARAG's design for two reasons. First, it explicitly recognizes "environmental and societal well-being" as a core trustworthy AI requirement, a dimension that is uniquely and directly instantiated in a carbon reporting system. Second, its requirements are sociotechnical rather than purely algorithmic, making them directly applicable to system-level design choices in agentic AI. As elaborated in Section 2, SMARAG operationalizes six of the seven EU HLEG requirements: human agency and oversight, technical robustness and safety, privacy and data governance, transparency, accountability, and environmental and societal well-being. The remaining requirement "diversity, non-discrimination and fairness" is acknowledged as an open challenge and is positioned as a priority direction for future research.

To overcome these challenges, we propose a TAI-powered carbon reporting system to provide accurate, efficient, and trustworthy carbon reporting, namely Self-decisive Multi-Agentic Retrieval-Augmented Generation (SMARAG, accessed with http://smarag.cee.cornell.edu), that integrates LLMs, multi-agentic system, and Retrieval-Augmented Generation (RAG). On the one hand, a multi-agentic system, as an intelligent system composed of multiple interacting agents, can be used to trigger a human-in-the loop workflow to execute specific tasks and meet stakeholder expectations (Liu et al., 2026; Lupoiu et al., 2025). By deploying multiple self-decisive agents to represent the key stakeholder roles (e.g., sustainability officers, project managers, consultants, and report writers) with distinct tasks and expectations in the workflow (Piccialli et al., 2025), the multi-agentic system can dynamically retrieve, optimize, and validate carbon report generation. On the other hand, RAG provides the domain and entity-specific knowledge base to ensure accurate, evidence-based, and context-aware carbon reports (Lewis et al., 2020; Siriwardhana et al., 2023). By implementing the RAG within privacy-protected secure infrastructure, sensitive data is handled or integrated before being sent to LLMs, to prevent information leaks while ensuring data fidelity for regulatory compliance. In this way, SMARAG is designed from the ground up as a trustworthy AI system, where each architectural decision corresponds to a specific EU HLEG requirement, as detailed in Section 2.

Here we present a practical roadmap to implement SMARAG as a trustworthy AI system for carbon reporting. First, guided by the EU Ethics Guidelines for Trustworthy AI, SMARAG operationalizes six of the seven EU TAI requirements, embedding them directly into architectural design. Second, we derive a standardized, stakeholder-aware carbon reporting workflow from real-world regulatory practices and professional sustainability reporting standards, which guides the agentic planning, decomposition, and generation process. Key stakeholder roles, including sustainability director, project manager, data analyst, and report writer, are formalized as autonomous agents that coordinate through a self-decisive validation loop and human-in-the-loop architecture to ensure the outputs meet regulatory standards before finalization. Third, we construct a proprietary climate knowledge repository by integrating real-world carbon disclosures, public carbon standards, and regulatory frameworks within secure local infrastructure. As an illustration, the repository incorporates carbon reports from over 700 higher education institutions between 2008 and 2023, covering

structured carbon goals, plans, actions, and historical disclosures to support domain-specific and regulation-aligned generation. Fourth, we deploy five state-of-the-art LLMs, such as GPT-4o, Claude 3.5, Gemini 1.5, Mistral-Instruct v0.3, and DeepSeek R1, as cognitive engines for each autonomous agent, operating through secure RAG pipelines within the system's privacy-preserving architecture. Finally, we validate SMARAG by generating carbon reports for 2017–2020 and benchmarking against three baselines: Multi-Agentic RAG (MARAG), single-agent (Vanilla) RAG, and standalone LLMs. Across five evaluation metrics, including experiment quality, report quality, usefulness, faithfulness, and relevancy (Durmus et al., 2020; Schmidgall et al., 2025; Zheng et al., 2023; Zhong et al., 2022), SMARAG consistently outperforms all baselines, with the performance gap between SMARAG and MARAG specifically isolating the incremental contribution of self-decisive task decomposition over standard multi-agentic RAG. These results collectively demonstrate that the combination of TAI-grounded system design, self-decisive decomposition, stakeholder-aware workflow formalization, and privacy-preserving knowledge architecture produces a qualitatively distinct system for knowledge-intensive regulatory reporting domains.

The rest of this paper is organized as follows. Section 2 establishes the SMARAG system design, covering the principles of trustworthy AI, the requirements and properties of the knowledge-intensive reporting domain, the system architecture, and the self-decisive multi-agentic carbon reporting workflow. Section 3 describes the system implementation, including the AI knowledge repository construction, multi-agentic system development, system deployment and prototype demonstration, and data security and privacy protection. Section 4 presents experimental validation, demonstrating SMARAG's effectiveness across multiple evaluation metrics and its superiority over baseline approaches. Section 5 concludes the paper and outlines directions for future research.

2. SMARAG system design

This section establishes the theoretical and design foundations of SMARAG as a trustworthy AI system, proceeding from guiding principles to architectural instantiation. Section 2.1 grounds the design in the EU Ethics Guidelines for Trustworthy AI, including the three-tier human oversight lifecycle. Section 2.2 characterizes the knowledge-intensive regulatory reporting domain and its five fundamental properties. Section 2.3 presents the system architecture design. Section 2.4 describes the self-decisive multi-agentic carbon reporting workflow.

2.1. Principles of trustworthy AI system design

The more AI applications are used in high consequence regulatory areas, the more crucial it becomes that they are designed in an accountable and trustworthy manner. The Ethics Guidelines for Trustworthy AI published by the European Commission High-Level Expert Group on Artificial Intelligence (EU HLEG) provide an internationally accepted normative framework to assist in this process of responsible development and deployment of an AI application. The framework defines Trustworthy AI through three interrelated dimensions: (1) it must be *lawful*, complying with applicable regulations and laws; (2) it must be *ethical*, adhering to ethical principles and values; and (3) it must be *robust*, both technically and socially, to ensure resilience across the AI lifecycle. These three dimensions are operationalized through seven requirements: human agency and oversight, technical robustness and safety, privacy and data governance, transparency, diversity and non-discrimination, societal and environmental well-being, and accountability.

SMARAG is designed to operationalize six of these seven TAI requirements within the domain of carbon reporting. Specifically, SMARAG addresses human agency and oversight through its stakeholder-grounded agentic structure; technical robustness through multi-LLM deployment and self-decisive task decomposition; privacy and data governance through its locally hosted, encrypted knowledge repository; transparency through interpretable agent reasoning and report traceability; societal and environmental well-being through its direct support for mandatory carbon disclosure; and accountability through structured inter-agent critique and reproducible reporting pipelines. The remaining requirement "diversity and non-discrimination" is not the primary focus of carbon reporting systems and is therefore outside the scope of the current study.

A central principle in trustworthy AI design is the appropriate allocation of human oversight across the AI system lifecycle. Kaur et al. (2022) propose a three-tier oversight model that distinguishes human roles by their temporal relationship to system decisions. Fig. 1 displays this model as it is adapted and extended to be appropriate in the SMARAG context and includes the following three tiers: (1) human-before-the-loop, (2) human-in-the-loop, and (3) human-on-the-loop. In the first tier, domain experts (e.g. sustainability director, project manager) and architects of the AI system establish requirements, workflow formalization, and design of the knowledge repository before any operation of the system takes place. Practically, two key documents will be produced at this stage: stakeholder requirement document and specification document. The stakeholder requirements document includes the key stakeholders and relationship analysis, along with the high-level requirements. The specification document is made to guide the developer to implement the TAI system with key principles, refined requirements and clear architecture. In the second tier, development and compliance officers of the AI system participate in all phases of development and validation through iterative refinement of data collection, model construction, and system integration. Finally, in the third tier, all the end-users of the AI system (i.e., sustainability officers, project managers, and consultants) interact with the system while performing live carbon reporting tasks by providing real-time oversight, applying judgement with respect to outputs created from the AI, and maintaining complete approval authority with respect to documents submitted for disclosure. This design approach of incorporating three levels of human oversight throughout the life of the SMARAG system provides a continuous and layered governance structure on human oversight rather than a singular point of intervention. Incorporating these principles of oversight into the design of the SMARAG system will ensure that SMARAG meets the requirement of the EU HLEG related to human agency and oversight related to the use of AI, at multiple levels, rather than as an ex-post compliance check.

2.2. Requirements and properties of Knowledge-Intensive reporting domain

Carbon reporting belongs to a broader class of domains that we term Knowledge-Intensive Reporting Domain (KIRD). Fig. 2 illustrates carbon reporting as a KIRD instance. On the left, measuring instruments and enterprise information systems (EIS) collect Scope 1, 2, and 3 emissions data through structured calculation pipelines. In the center, a six-step regulatory workflow, from establishing a reporting group to final report writing, transforms that data into a structured carbon report. On the right, the report is disclosed to a multi-stakeholder environment comprising government regulators, rating agencies, stock exchanges, sustainability consultants, and the public. This end-to-end structure reveals five domain properties below:

Property 1 — **Regulatory Compliance Dependency (P1-RCD).** In KIRD, the structure, scope, and content of reports are governed by external, versioned regulatory standards that carry legal or reputational consequences for non-compliance. In the carbon reporting context, these standards include the GHG (Greenhouse Gas) Protocol, Climate Disclosure Project (CDP, 2024) Corporate Reporting Framework, and ISO (The International Organization for Standardization) 14064, each with distinct boundary definitions, calculation methodologies, and disclosure formats (Howard et al., 2015). Compliance must be dynamically

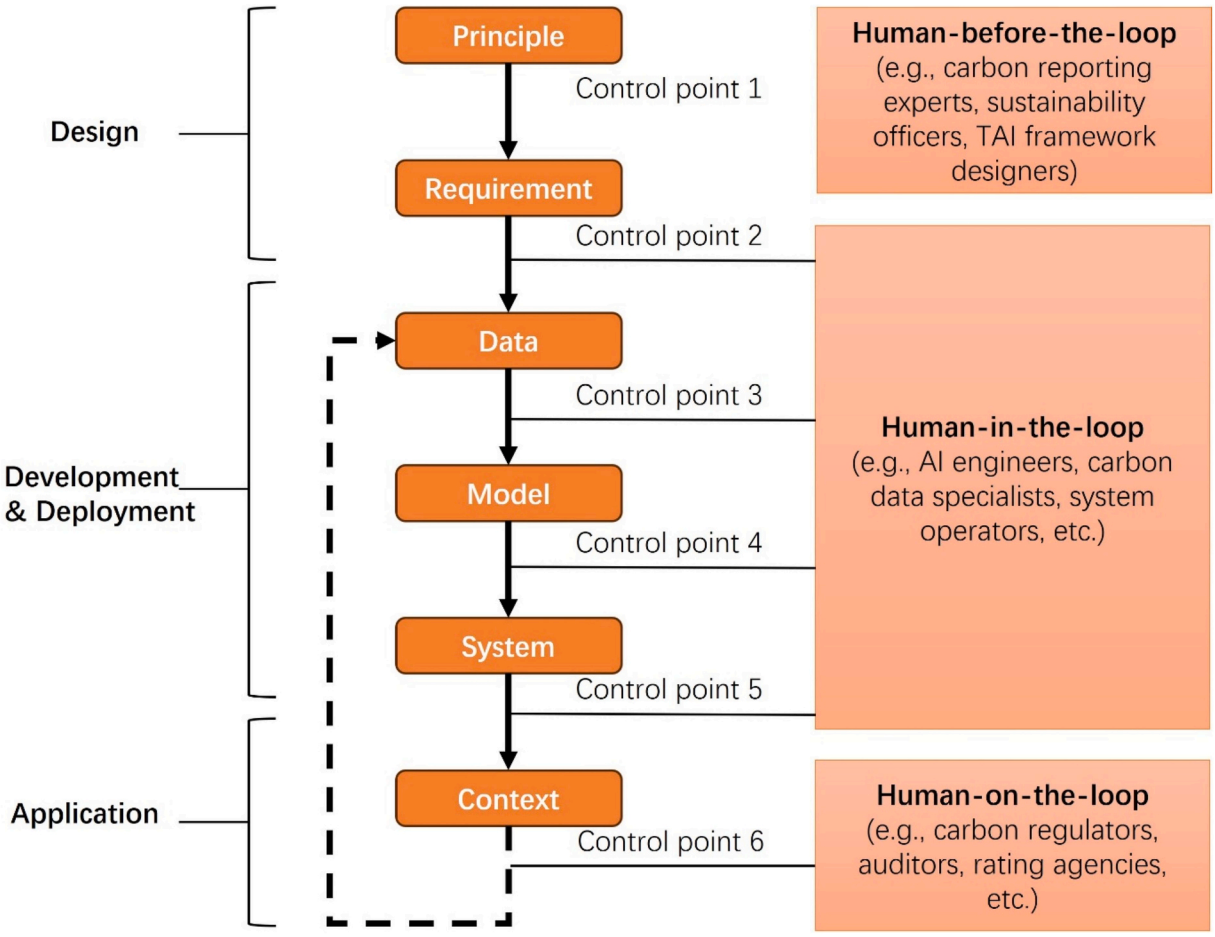


Fig. 1. The practical roadmap to implement a trustworthy AI system, inspired by (Kaur et al., 2022).

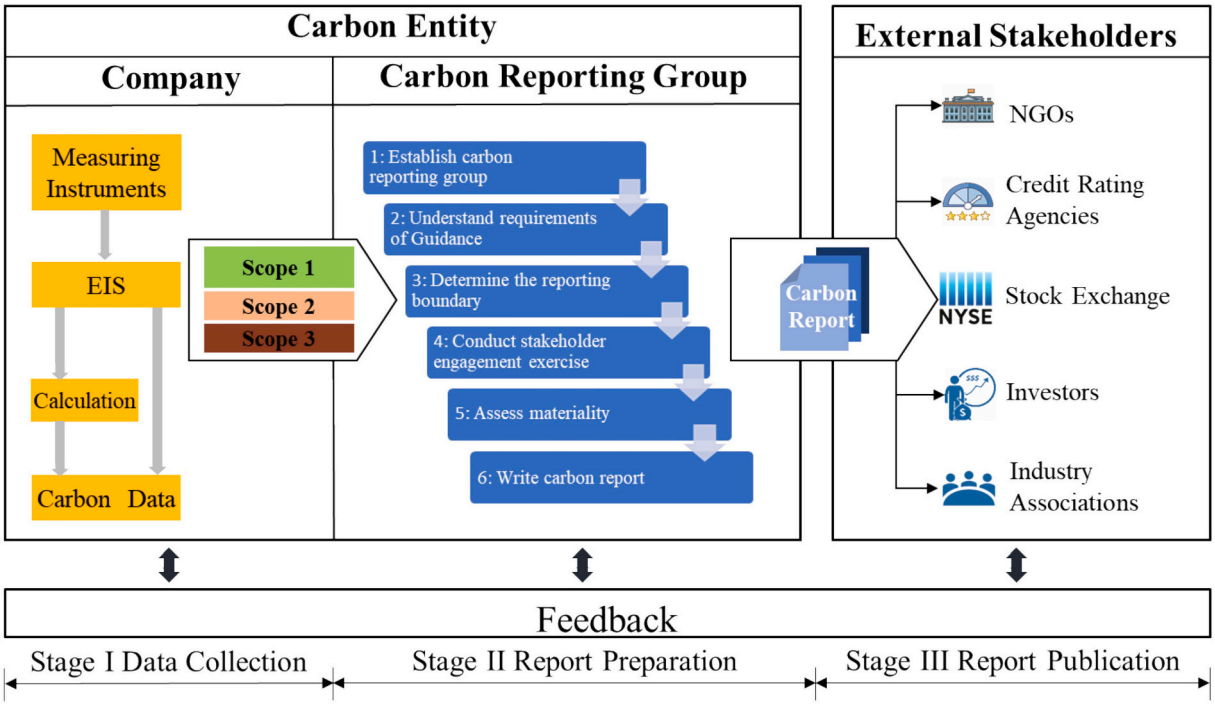


Fig. 2. The carbon reporting workflow in knowledge-intensive reporting domain.

maintained as standards evolve. This property drives the need for a dedicated standard layer with versioning support, standard-aware retrieval, and compliance validation embedded in each generation step. *Analogous domains include* financial reporting, ESG disclosure, etc.

Property 2 — **Multi-Stakeholder Workflow Structure (P2-MSWS).** KIRD is not a single-entity task. On the one hand, report generation involves a structured chain of organizational roles. In carbon reporting, these roles include sustainability director (scope definition), project managers (workflow oversight), data analysts (emissions data), consultants (analysis and computation), and report writers (documentation) (Zhou, 2020). On the other hand, Scope 3 disclosures require data from the upstream and downstream partners in supply chains. Therefore, this property drives the multi-agentic design of SMARAG, in which autonomous agents are instantiated with role-specific task definitions, evaluation criteria, and inter-agent communication protocols, coordinated by the self-decisive task breakdown engine.

Property 3 — **Proprietary Data Sensitivity (P3-PDS).** KIRD involves commercially or legally sensitive organizational information that cannot be transmitted to external AI systems. In carbon reporting, the data, such as fine-grained emissions records, production process data, and electricity consumption data, constitutes business secrets and proprietary assets (Preuss, Alshehri and You, 2024; Yao et al., 2024). This property drives the three-tier knowledge architecture and local RAG deployment: all proprietary data is processed on-premises, and only

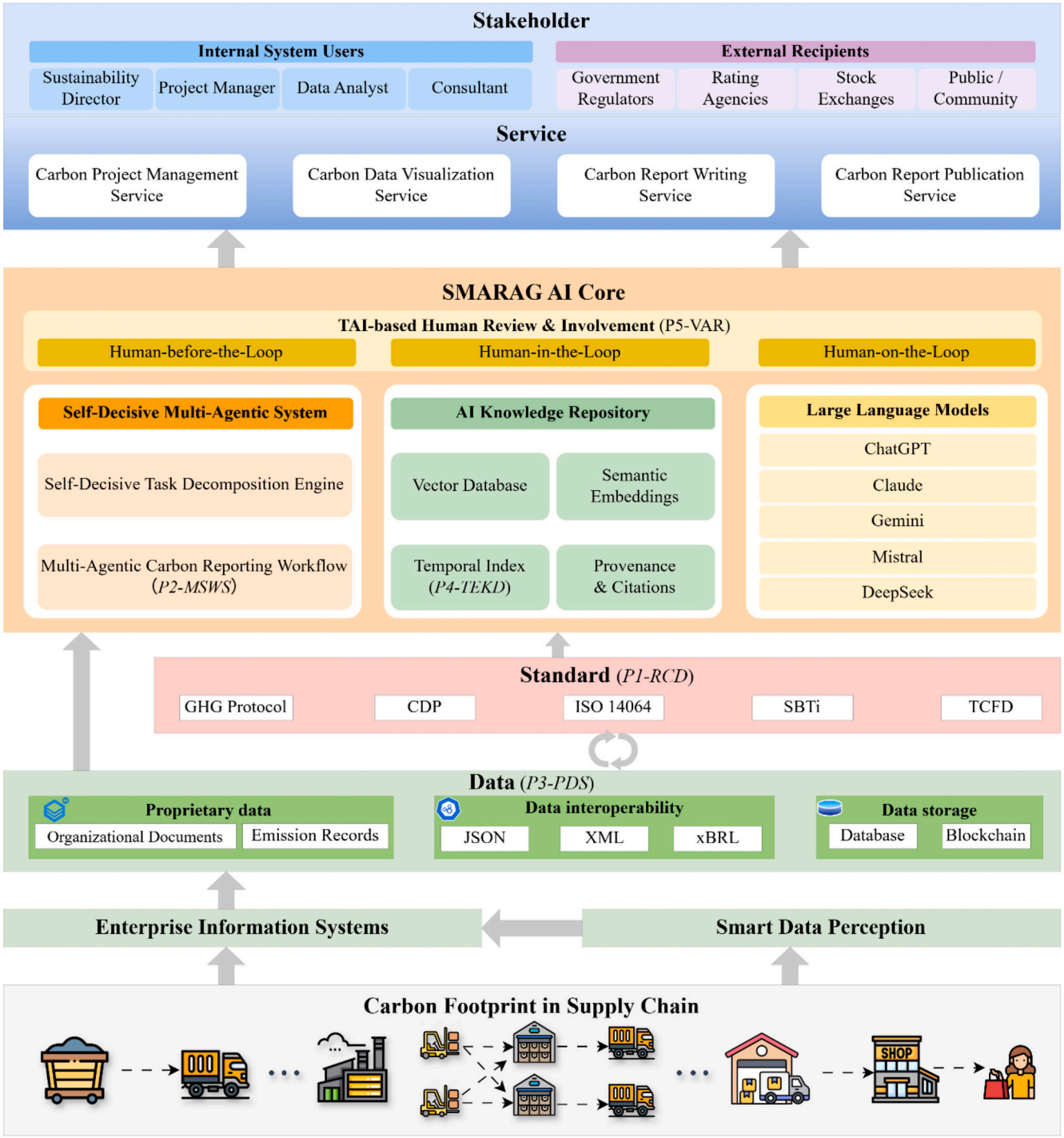


Fig. 3. The system architecture of TAI-powered carbon reporting.

contextually filtered passage-level snippets are transmitted to external LLM APIs, minimizing data leakage risks.

Property 4 — **Temporal-Evolutionary Knowledge Dependency (P4-TEKD).** In carbon reporting, both the knowledge base and the regulatory environment evolve over time. Reports require historical data for trend analysis, target-setting, and year-over-year comparability, while the standards against which they are evaluated are periodically updated. This property drives the real-time tracking of public standards, the historical data repository spanning multiple reporting years, and the temporal dimension of SMARAG's experimental validation (2017–2020).

Property 5 — **Verifiability and Auditability Requirement (P5-VAR).** The generated reports must be traceable to source data, defensible under external scrutiny, and free from unsupported claims. AI hallucinations are not merely inaccurate but potentially legally or regulatorily consequential. In carbon reporting, third-party auditors and regulatory bodies verify disclosed emissions data against source records (Bellassen et al., 2015). This property drives the faithfulness-centric evaluation metrics, the iterative inter-agent critique loops, the human-in-the-loop verification gates at each workflow stage, and the fact-checking mechanism embedded in each agent execution step of SMARAG.

Understanding these domain properties explicitly serves two purposes. First, it grounds SMARAG's design choices as principled responses to formal requirements. Second, it identifies the set of domains to which SMARAG's architecture can be meaningfully adapted. Moreover, the five KIRD properties correspond directly to six of the seven EU HLEG TAI requirements introduced in *Section 2.1*. This mapping demonstrates that SMARAG is a principled, requirements-driven architecture in which each trustworthy AI requirement is traceable to a concrete design decision. The detailed system architecture with requirements-to-design mapping is presented in *Section 2.3*.

2.3. System architecture

SMARAG is grounded in a seven-layer architecture designed to operationalize TAI principles across the full carbon reporting lifecycle. As shown in Fig. 3, the layers span from the physical supply chain scenario at the bottom to the human stakeholder at the top, with the SMARAG AI Core at the center. Each layer tackles certain KIRD properties and EU HLEG TAI requirements, ensuring that transparency, accountability, and human oversight are embedded throughout the system.

The **scenario layer** presents a typical carbon footprint scenario in the supply chain from raw material production, transportation and logistics, manufacturing, distribution, sales, and consumption. Each process generates the carbon footprint, and the carbon entity can be any stakeholder in the supply chain. Carbon emission data can be collected in this layer for further carbon reporting and analysis. Directly above the scenario layer, **enterprise information systems** and **smart data perception** provide the data acquisition infrastructure. Carbon data are collected from smart data perception and stored in enterprise information systems, such as Enterprise Resource Planning, Supply Chain Management, Human Resource System, etc. Enterprise systems supply structured organizational data, feeding the data layer above.

The **data layer** (P3-PDS) manages all carbon-relevant data assets within SMARAG. It organizes information into three modules: proprietary data, data interoperability, and data storage. The proprietary data comprises organizational documents and emission records, serves as the organization-specific knowledge base. The data interoperability methods include JSON, XML, and xBRL, to facilitate data exchange among different users. The data storage approach can be database or blockchain depending on the data transparency requirements of the carbon entity.

With the carbon data ready, carbon standards are needed to generate the carbon report. The **standard layer** (P1-RCD) encodes the mainstream carbon reporting frameworks that govern disclosure obligations. It includes the existing mainstream carbon reporting standards, such as GHG Protocol, CDP, ISO 14064, etc. These standards are maintained with a version-tracking mechanism, directly informing both the data layer below and the SMARAG AI Core above.

The **SMARAG AI Core** is a system engine, integrating three interdependent components under a unified TAI-based Human Review and Involvement governance framework (P5-VAR). The Self-Decisive Multi-Agentic System includes the self-decisive task decomposition engine and the multi-agentic carbon reporting workflow (P2-MSWS), which orchestrates automated report generation across parallel agent streams. The detailed construction of self-decisive multi-agentic carbon reporting can be checked in Section 2.4. The AI Knowledge Repository grounds all generation in verified knowledge, comprising a Vector Database for RAG-based retrieval, Semantic Embeddings for contextual matching, a Temporal Index (P4-TEKD) to ensure temporal consistency of cited standards and data, and Provenance and Citations tracking to maintain complete audit trails of all retrieved content. The detailed AI knowledge repository construction can be checked in Section 3.1. The LLMs (e.g., ChatGPT, Claude, Gemini, Mistral, DeepSeek) that serve as our generative backbone. Different LLMs are integrated with SMARAG to increase the system's robustness and identify suitable LLM tools. All three AI Core components operate under a three-tier human oversight structure that operationalizes requirement P5-VAR. *Human-before-the-Loop* establishes governance constraints and system design parameters prior to design and development, encoding human principles and ethical boundaries into the system architecture. *Human-in-the-Loop* enables direct human intervention during report generation, allowing validators to review, correct, and approve AI-generated carbon content at critical checkpoints. *Human-on-the-Loop* provides continuous supervisory monitoring with the authority to override AI decisions, ensuring that human accountability is preserved across the full reporting workflow.

The **service layer** translates AI Core outputs into actionable carbon reporting services. There are four types of services: carbon project management service, carbon data visualization service, carbon report writing service, and carbon report publication service. Specifically, project management service is used by the manager to create, configure, and manage the carbon reporting project. Carbon data visualization service is used by managers or data collectors to check the carbon reporting progress. Carbon report writing service is used by the consultant to write the carbon report. Carbon report publication service is used for carbon entity to publish the carbon report to relevant regulators or rating agencies.

The **stakeholder layer** encompasses all human actors engaged with the SMARAG system, partitioned into two groups reflecting their distinct roles in the carbon reporting process. Internal system users, such as sustainability directors, project managers, data analysts, and consultants, interact directly with the service layer to initiate, review, and approve carbon reporting workflows. External recipients, such as government regulators, rating agencies, stock exchanges, and the public/community, receive the finalized Carbon Report as the system's primary output, fulfilling the disclosure obligations encoded in the standard layer (P1-RCD). This explicit partitioning reflects the TAI requirements of accountability and transparency, ensuring that outputs are traceable to appropriate stakeholders and that the system's reporting chain is auditable end-to-end.

2.4. Self-Decisive Multi-Agentic carbon reporting workflow

As shown in Fig. 4, we propose a Self-Decisive Multi-Agentic RAG-based approach to automate the process of generating carbon reports and their analysis based on the historical information of the carbon entity. In the middle of Fig. 4 is the core of our approach. The user's prompts are used in RAG, task formulation and multi-agentic system. In RAG, relevant carbon standards, historical data, and documents are searched, analyzed, and integrated with the user's prompts. Then all the

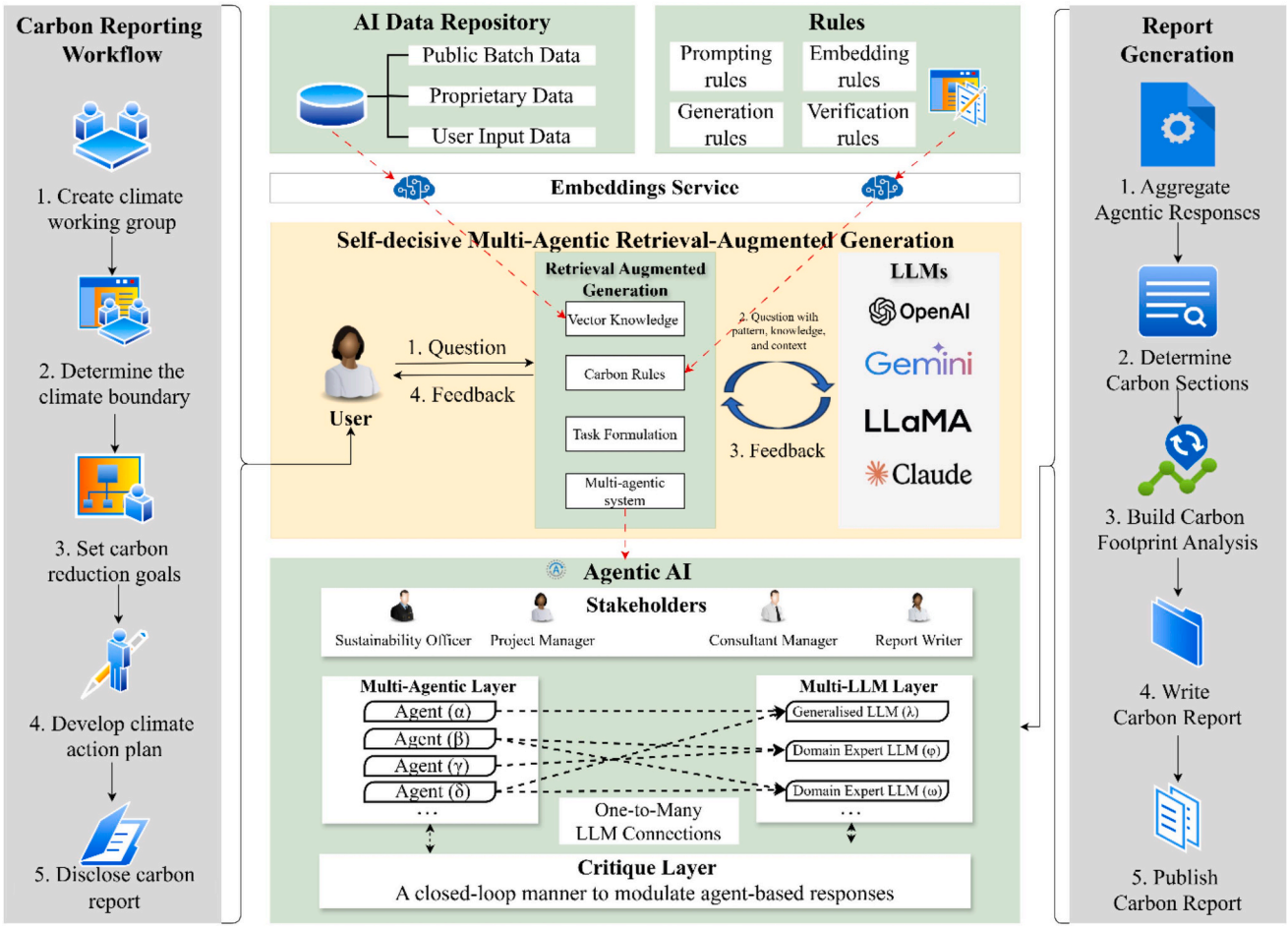


Fig. 4. Self-Decisive Multi-Agentic RAG-based approach for carbon reporting.

integrated information will be first sent to self-decisive task breakdown engine, where we plan and divide the whole user request into multiple subtasks by carefully analyzing each part of the request and augmenting it with the information retrieved from RAG. Each task is rigorously validated based on a set of Carbon Rules and Criteria to check its effectiveness.

Once validated, we create multiple task-specific prompts (including task information, carbon sections, retrieved relevant information, etc.) and invoke LLMs, to generate each task-level text. Each Agent, when required (for example, when a large carbon section is divided into smaller sub-sections that are being generated by different agents), communicates and shares context among themselves to produce more accurate text. We also check for factuality against the data stored in our knowledge store and user input at each Agent execution to validate the responses. Finally, when all task-level texts are generated, we combine all of them together to form a high-quality carbon report. We further perform evaluations and revisions at a report level to validate all the required information.

At the top of Fig. 4 are AI data repository and rules. In our developed SMARAG, we define three data types in AI data repository: public batch data, proprietary data, and user input data. Public batch data consists of carbon standards, good practices and previous carbon reports of the organizations that are key drivers of the required context for our Agents. However, this data is not static, so we keep tracking it in real time for better knowledge grounding. Proprietary data is organization- and department-specific private information that cannot be obtained on the Internet, so we've surveyed various organizations to obtain the data as part of our research. User input data specifies the necessary constructs of the primary aspects required for the report. Rules refer to different applying rules in the RAG such as prompting rules, retrieval rules, etc.

At the left side of Fig. 4 is the carbon reporting workflow generalized from the real-life carbon reporting practice based on the EPA Simplified Guide to Greenhouse Gas Management for Organizations. It presents the workflow requirements, stakeholder expectations, and evaluation criteria, which are useful to constructing multiple agents, such as sustainability officers, project managers, consultant managers, and report writers, etc.

At the bottom of Fig. 4 is the agentic AI. Our Self-Decisive Multi-Agentic AI Approach presents a unique process of prompt construction and prompt revision in an automated fashion by leveraging the prior context along with stakeholder expectations. Each Agent may interact with one or more Domain-Expert and Generalized LLMs for action planning and response modelling while utilizing the Tools of API calling or database query generation wherever required. Further, our Critique layer revises the constructed prompts to filter the generated responses to ensure fairness, grounded facts, and reduce hallucinations.

The report generation workflow in RAG is at the right side of Fig. 4. This is reached upon completion of multiple stages of generated responses, prompt/context revisions through agentic critiques, fact checking and hallucination handling corresponding to each section/requirement. Now, our Agentic Response Aggregator aggregates all pieces of text based on the user-selected carbon standard and determines the required carbon sections for the report.

Additionally, as a final step, we generate carbon footprint analysis to be added as part of the report. We perform this at the end because we now have accurate information about the company's carbon data which is then added as part of relevant sections before publishing the report.

3. System implementation

3.1. AI knowledge repository construction

To clarify the theoretical construction of AI knowledge repository in SMARAG, we organized the processes into three steps, including (1) **data integration**, (2) **normalization**, and (3) **semantic clustering**.

The first step is data integration. All relevant data sources, including public datasets (e.g., carbon standards and best practices), proprietary information (e.g., internal carbon goals and emissions data), and user-provided contextual inputs, are preprocessed and transformed into high-dimensional vector representations using a dedicated embedding service. These embeddings serve as the foundational layer for semantic retrieval, allowing the system to match queries with contextually relevant and regulation-aligned information across heterogeneous data domains. This unified representation is critical for grounding agentic reasoning and ensuring that subsequent report generation is both accurate and domain specific.

The second step is data normalization. The data are normalized using the L2 norm before feeding them into the vector database (Yu et al., 2010). The normalization is essentially applied over each of the carbon embedding (CE) Vector independently, shown in **Equation (1)**:

$$\|CE\|_2 = \sqrt{\sum_{i=1}^{n} |CE_i|^2} \tag{1}$$

Each of these normalized embeddings is then stored into the vector database. Based on the carbon entity and stakeholder information, there could be various related information available in the database.

The third step in the SMARAG workflow is semantic clustering. We identify the best information by forming clusters of them and ranking each of these clusters according to the formed contextual plan. We chose the Davies-Bouldin Index (Ros, Riad and Guillaume, 2023) to compute the number of clusters primarily because it captures both the compactness and separation of the clusters. To suit our requirements for performing the search over the documents based on the carbon section along with the formed contextual action plan and to refine the similarity matching for better grounding of our results, we use the following modification of the core formula in **Equation (2)**:

$$S_i = \sqrt{\frac{1}{T_i} \sum_{j=1}^{T_i} \|CS_j - A_i\|_2^2} \tag{2}$$

The Standard deviation (S_i) between the features in clusters is obtained by using CS_j as the carbon section embeddings, we choose to have the dispersion measure (q) to be the standard deviation with the distance metric (p) as the standard Euclidean Distance performed over all the Centroids (A) of the Cluster having size as T_i.

We also modify the formulation the Measure of separation ($M_{i,j}$) between the sections to be computed based on the Carbon Embeddings ($CE_{k,I}$ and $CE_{k,j}$ corresponding to each cluster i and j). This is shown in **Equation (3)**.

$$M_{ij} = \sqrt{\sum_{k=1}^{n} |CE_{k,i} - CE_{k,j}|^2} \tag{3}$$

Then, in **Equation (4)**, we compute the Davies-Bouldin Index to find the optimal number of clusters required to classify the current plan to and to obtain the relevant contexts.

$$DB = \frac{1}{N} \sum_{i=1}^{N} \max_{i \neq j} \left(\frac{S_i + S_j}{M_{ij}} \right) \tag{4}$$

We further use Cosine Similarity (S_{cos}) between the Carbon Embedding (CaE) and the Contextual Embeddings (CtE) as a measure to find the most relevant contexts in the predicted cluster for all our three different types of data (Public (Pu), Proprietary (Pr) and User Input (Ui) Embeddings). This statistic appears in **Equation (5)**. Then, we perform the Contextual Summarization (CTX_{summ}) on the Cosine Similarity to form the response (S_{res}) for the Carbon Section grounded on the required data, shown in **Equation (6)**. Essentially, we take the union of all the similar contexts, one from each type of data.

$$S_{cos}(CaE|\{Pu, Pr, Ui\}) = \bigcup_{i \in \{Pu,Pr,Ui\}} \frac{\sum_{j=1}^{n} CaE_j^i \bullet CtE_j^i}{\sqrt{\sum_{j=1}^{n} CaE_j^i} \bullet \sqrt{\sum_{j=1}^{n} CtE_j^i}} \tag{5}$$

$$S_{res}(R|\{Pu, Pr, UI\}) = CTX_{summ}\langle S_{cos}(CaE|\{Pu, Pr, Ui\})\rangle \tag{6}$$

The Contextual Summation statistic CTX_{summ} is based on the Rules (R) and the stakeholder-specific instructions and for evaluating the retrieved contexts and formed summary against these requirements, we use the ROUGE (Recall-Oriented Understudy for Gisting Evaluation) as a metric for evaluating the generated texts.

3.2. Multi-Agentic system development

SMARAG employs the paradigm of a service-oriented architecture (SOA), which is adapted for two key reasons. First, SOA's key features of loose coupling and replaceability mean that components (e.g., LLM providers, agent policies, or data engines) can be independently versioned or replaced without regard to the end services, allowing for scientific reproducibility and fast iterations (Papazoglou and Georgakopoulos, 2003). Second, independent scaling and fault isolation provide reliability and utilization. For example, elastic Celery workers can handle spikes in workload, long-running AI processes can be run independently from handling requests, and if a failure occurs, it will remain localized. Such configurations enhance capabilities such as latency-tolerant large-scale services and queue-based architecture. Collectively, these attributes provide the maintainability, portability, and agility required for a research-to-production system that must adapt as the LLM, datasets, and workflows evolve.

The modularity of the SMARAG system capabilities is established using loosely coupled services structured in five layers, as shown in Fig. 5. The Frontend Layer utilizes a React client and a WebSocket Context/Global Connection Manager, maintaining the session state, and separates presentation logic from transport logic. The backend layer implements the articulated interfaces contract-first with FastAPI Backend (HTTP and WebSocket endpoints) and is coordinated by the WebSocket Manager/Connection Handling service to maintain long-lived, bi-directional sessions. This establishes the separation of concerns, producing a client that is thin and stateless, while guaranteeing security, validation, and orchestrated semantics for the backend service contracts. The analytics components are assembled in the AI Layer, which includes a Multi-agentic System (Planning & Generation), Celery Workers (Background Tasks) for scalable task activation, and LLMs (e.g., ChatGPT, DeepSeek, etc.). The Data Layer encapsulates persistent and high-throughput data operations. For instance, we use MongoDB (Primary Database) to be the system of record, Redis (Cache & Message Broker) for low-latency cached and asynchronous command/response, staged event-driven architectures, and Milvus Vector DB (Embeddings & RAG) with embeddings for retrieval-augmented generation (Lewis et al., 2020). Cross-cutting operational concerns are encapsulated as the Monitoring Layer, where Flower provides observability for Celery pipelines, and Attu UI provides administration and health monitoring for Milvus indexes. Inter-service communication occurs exclusively using explicit APIs and schemas for messages, thus avoiding direct database coupling so that schemas may continue to evolve post-implementation. Technical details and parameters can be seen in Table 1.

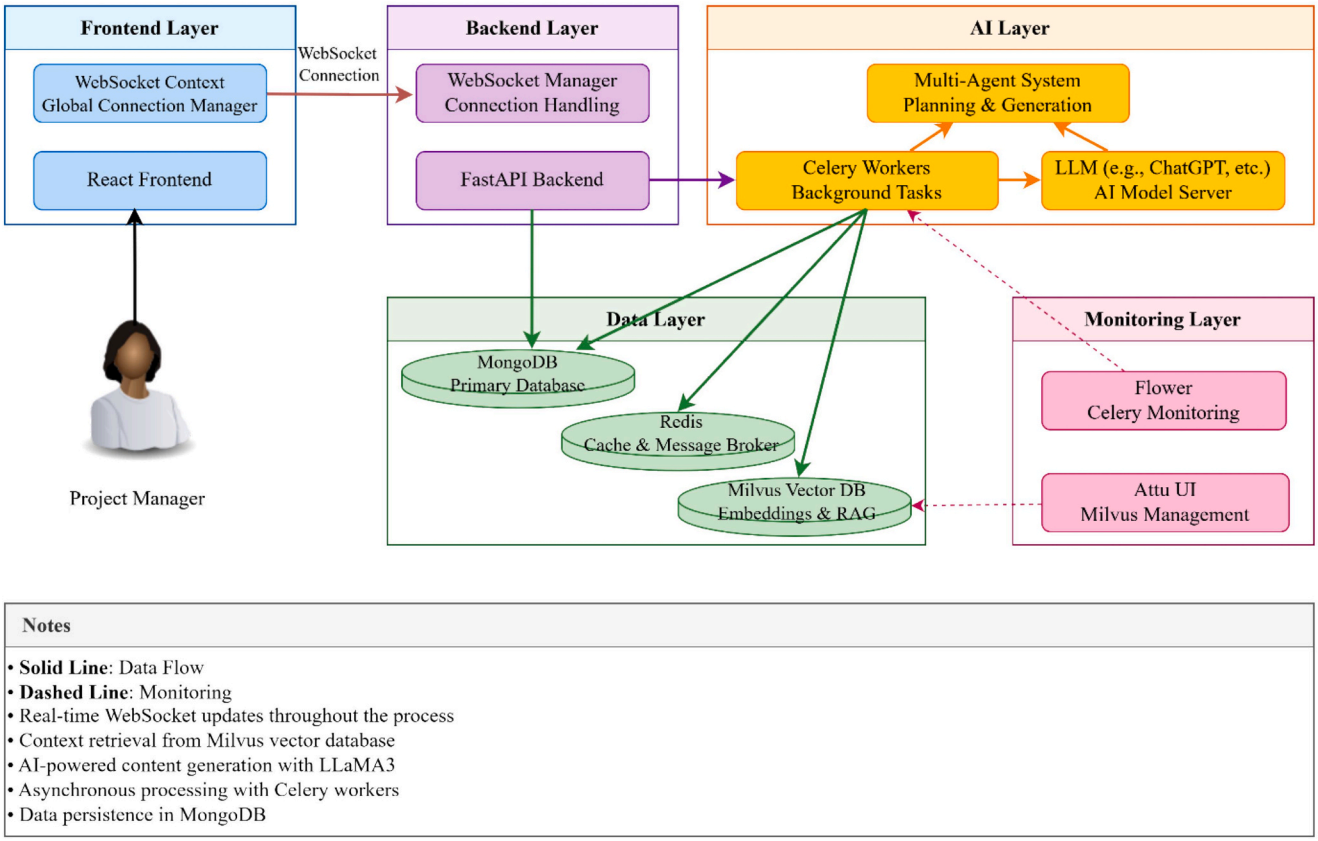


Fig. 5. Service-oriented design and development for the SMARAG system.

Table 1
Development tool and environment for SMARAG.

Component		Description
Integrated Development Environment		Visual Studio Code
Front-end programing language		HTML, CSS, React.js
Backend component	Backend API	FastAPI backend with Swagger docs
	Agent creation	Python 3.10
	Data cache	Redis 7.2.x
	Distributed task queue system	Celery 5.5.2
	Database	MongoDB 7.0.x
	Vector database management	Milvus Attu UI
	Local LLM inference	ollama-llama3.2
Operating system		macOS Sequoia 15.6.1

3.3. System deployment and prototype demonstration

We provide below the deployment environment for SMARAG, which is summarized in Table 2. SMARAG is running on a single Ubuntu 24.04.3 LTS server with 8 vCPUs (Intel Xeon Gold 6354 @ 3.00 GHz), 31 GiB RAM, and 4 GiB swap. Application data is stored persistently on a 400 GiB SSD data volume, while the operating system and runtime libraries are stored on the system disk. This provides a balanced CPU–memory footprint suitable for sustained inference, retrieval, and preprocessing workloads common to our pipeline. The entire software stack is containerized using Docker with Compose to allow for reproducible builds and easy orchestration of SMARAG services. External access to SMARAG is proxied by Nginx, acting as a reverse proxy, terminating HTTP(S) on ports 80/443 and proxying connections to the internal service endpoints. This isolates application concerns (service composition and scaling) from edge concerns (request processing and protocol termination), while limiting the overall deployment to a single host, and keeping the deployment simple and easy to maintain.

Table 2
Server parameters for SMARAG deployment.

Item	Parameter
CPU cores	8 vCPUs (Intel Xeon Gold 6354 @ 3.00 GHz)
RAM	31 GiB total
Swap	4 GiB
OS	Ubuntu 24.04.3 LTS
Data disk	400 GiB
Nginx	Reverse proxy tool

Fig. 6 illustrates how the interactive human-AI workflow of SMARAG generates a carbon report. After signing up and authenticating, the user inputs carbon data to a structured input form (e.g., standards, goals, plans, and actions) and can track report-generating history for provenance. Then the SMARAG generates a generation plan documenting the sections that will be showing up in the coming carbon report. The generation plan is presented to the user for verification to ensure that report sections are generated as intended. If the generation plan is not acceptable, the user can change their inputs or request a new generation plan by giving the new prompts through the chat box. The verification gate is a strategy to stop unnecessary computation in future generation steps and help to formally establish an auditable intent for carbon report generations.

With the approval, SMARAG proceeds with executing the generation plan and produces a draft report (in preview mode). The user evaluates the report against internal criteria (e.g., completeness, factual conformity with input, consistency with standards) and can edit the report either as an AI-assisted edit (e.g., provides a structured prompt to specify or elaborate on a section or via manual edit (e.g., allows the User to just change text directly for more granular changes). The appraisal/editing process is repeatable until a generated draft satisfies the user's quality

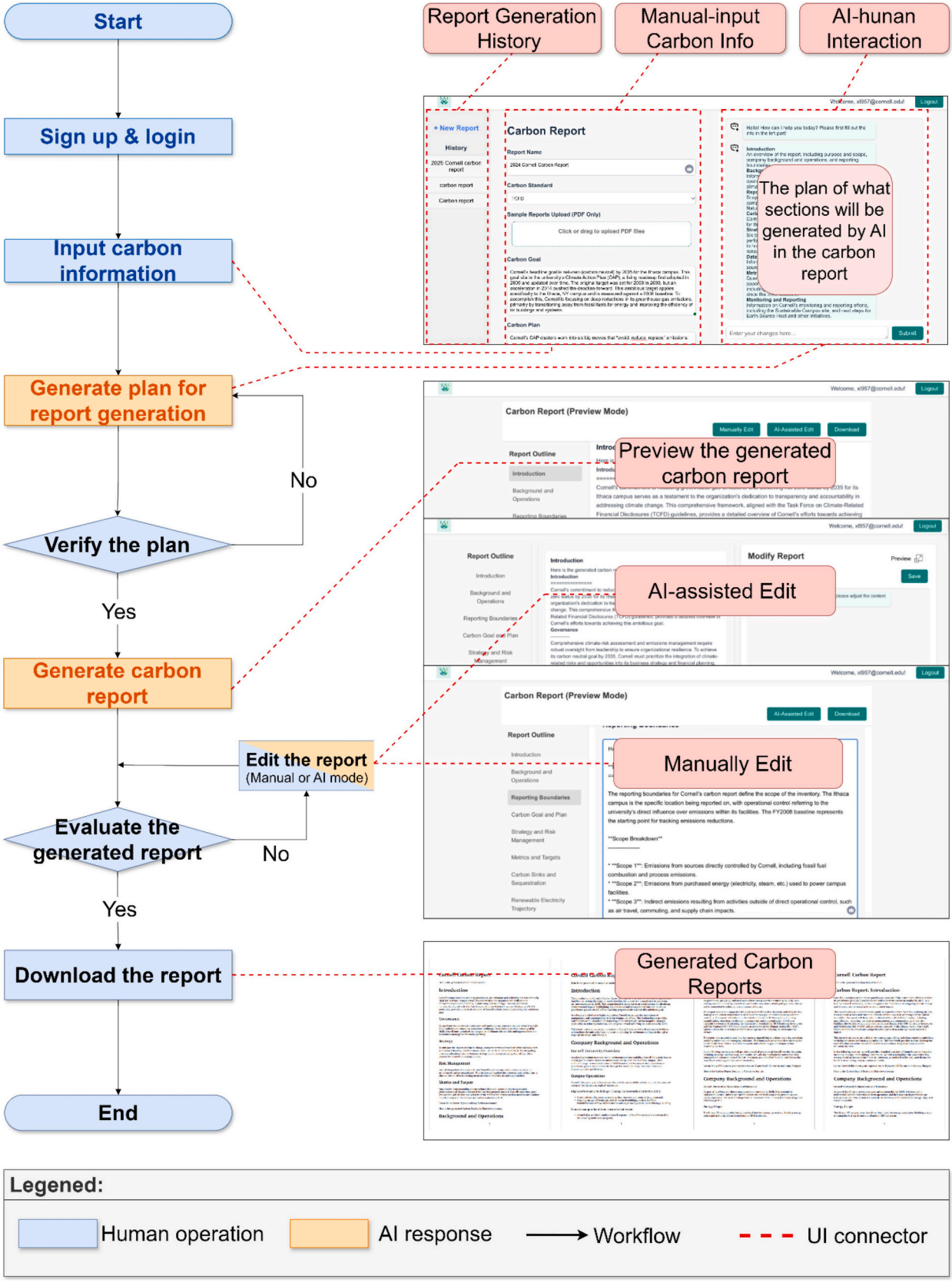


Fig. 6. The demonstration of generating carbon report using SMARAG.

expectations. All user interactions are available for the history, allowing the user to trace their inputs and choices later.

Once the report is approved by the user through appraisal, they then finalize and download the report to create the final report. The combination of verifying the plan, iterative previewing, and prompt-based drafting delivers the carbon report generation that is traceable, reversible, and grants control over substantive content. In practice, the loop reduces the time to precision-processing a high-quality report and extends the institution's governance of that content for accountability to external stakeholders.

3.4. Data security and privacy protection

Data privacy and security are foundational requirements for a trustworthy AI system in sensitive domains. The EU HLEG (2019) identifies "privacy and data governance" as one of seven core requirements for trustworthy AI, stipulating that AI systems must achieve confidentiality, data integrity, and legitimate data governance throughout the entire data lifecycle. For carbon reporting, where organizations must share proprietary emissions records to obtain regulatory guidance, this requirement requires particular urgency. SMARAG addresses it in three layers, as illustrated below.

At the data layer, SMARAG follows the data locality rule for all proprietary carbon data. The vector DB (Milvus) and primary document DB (MongoDB) are hosted exclusively on the organization's local servers and cannot be accessed from the public internet. During RAG-based report generation, the retrieval pipeline only pulls the most contextually relevant passages from the local knowledge store. Then these passage-level snippets are concatenated with a structured user query and sent to the external LLM API. Thus, even in the event of an LLM API breach, only contextually filtered fragments could be exposed, not the sensitive information or business secrets.

At the network layer, all the communications between external users and SMARAG are secured through HTTPS/TLS (HTTP on port 80 is redirected to HTTPS on port 443), with SSL/TLS termination performed by the Nginx reverse proxy. Acting as a security gateway, Nginx receives and authenticates all incoming requests and forwards them to the corresponding internal service endpoints (FastAPI backend, MongoDB, Redis, Milvus). All communications between each microservice are done exclusively within an isolated Docker bridge network which is not reachable from the public internet. This means that no database or AI service endpoint is directly exposed to the external world, vastly decreasing the attack surface.

At the infrastructure layer, SMARAG is currently hosted on institutionally secured cloud server, which operates under institutional data governance policies, physical security controls, and OS-level access restrictions (Ubuntu 24.04.3 LTS with role-based user privileges). User-level authentication is required before accessing any system function. The entire software stack is containerized with Docker Compose, providing service-level isolation: a failure or compromise in one container does not propagate to other services. Redis is used for session caching and task queuing (ephemeral, non-persistent by default), further limiting the persistence of sensitive in-flight data.

Collectively, data locality (data layer), encrypted transmission (network layer), and institutional access control (infrastructure layer) operationalize the EU HLEG "privacy and data governance" requirement at three distinct system levels, providing empirical grounding for one of SMARAG's trustworthy AI claims. Together, these three layers implement a comprehensive, regulation-conscious approach to data protection that directly addresses the privacy and security challenges identified in Section 1, ensuring that carbon entities can leverage TAI-powered reporting without compromising their proprietary information.

4. Experimental validation

4.1. Data sources

The data used for illustration in this study are from *Second Nature*, a public website committed to accelerating climate action in and throughout higher education, with a network of over 700 member institutions. The member educational institutions must collect, integrate, and report their emissions data to Second Nature in support of their campus climate action goals. We collected research data from Second Nature member institutions, including emissions data from 2008 to 2023. Based on their historical carbon data and reports, we built the AI-based knowledge repository, which in turn is used by our proposed SMARAG carbon reporting engine to generate four annual carbon reports for the years 2017–2020 using the existing carbon data of the universities based on the Carbon Disclosure Project (CDP). CDP is a non-profit organization that runs the widely-adopted global environmental disclosure system (CDP, 2024). As presented below, the proposed SMARAG approach significantly enhances CDP carbon reporting by automating data retrieval, ensuring compliance with CDP requirements, and improving the credibility and transparency of carbon disclosure.

4.2. Trustworthy AI-based evaluation metrics

Evaluating SMARAG requires addressing two distinct but complementary objectives: demonstrating carbon reporting effectiveness and providing empirical grounding for the TAI compliance claims established in Section 2.1. Accordingly, this section introduces five evaluation metrics that serve a dual purpose, measuring carbon reporting performance and simultaneously operationalizing measurable proxies for four of the six addressed EU HLEG TAI requirements. The remaining two addressed requirements are validated through architectural design evidence in Sections 3.4 and 4.4 respectively, as they do not admit direct quantitative operationalization within a single-system experiment.

Five critical evaluation criteria are identified from the literature (Durmus et al., 2020; Schmidgall et al., 2025; Zheng et al., 2023; Zhong et al., 2022) and used to conduct performance evaluation. Each criterion is measured on a scale from 0 (lowest) to 10 (highest): (1) **Experiment Quality**, (2) **Report Quality**, (3) **Usefulness**, (4) **Faithfulness**, and (5) **Relevancy**. These criteria reflect both process integrity and content validity. These metrics incorporate ROUGE scores (Lin, 2004) for Usefulness, context recall and precision for Report Quality (Yu et al., 2024), factual adherence for Faithfulness (Durmus et al., 2020), topic alignment for Relevancy, and semantic similarity for Experiment Quality (Zhang et al., 2019; Huang et al., 2026). Table 3 maps each metric to the EU HLEG TAI requirement it operationalizes, establishing the explicit evidential chain. For requirements addressed by quantitative metrics, the experimental results in Section 4.3 constitute direct empirical evidence of TAI compliance.

To measure **Usefulness**, we take the mean of three sub-indicators, each rescaled to a 0-to-10 scale, borrowing from ROUGE, a decades-old, well-respected package for automated measurement of text quality, improved in recent years (Ganesan, 2018; Lin, 2004). $\mathbf{ROUGE_1}$ describes how often the AI mentions the same key terms as the reference text, using the standard measure unigram overlaps, as shown in **Equation (7)**. $\mathbf{ROUGE_2}$ describes how often the AI selects the same word pairs (bigrams) as the reference text, showing that the AI is correctly associating concepts, as shown in **Equation (8)**. Finally, $\mathbf{ROUGE_{lcs}}$ evaluates the structure and coherence of the response, measuring how much the AI uses the same longest common subsequences as the reference text, as shown in **Equation (9)**. This means that the AI preserves the same linguistic style, structure, and logical flow as the original.

- $\mathbf{ROUGE_1}$ (unigram overlap):

Table 3
The evaluation metrics for trustworthy AI system.

Metric	Measurement Approach	EU TAI Requirement	Role in TAI Evidence
Usefulness	ROUGE score against reference carbon reports (Eq. (7)–(9)	R1: Human Agency & Oversight; R7: Accountability	Measures whether AI-generated drafts are actionable for human reviewers, supporting effective Human-in-the-Loop appraisal and final approval
Report Quality	Context recall and precision against carbon standard content (Eq. (10)–(11)	R4: Transparency	Measures completeness and accuracy of knowledge retrieval; high scores confirm that outputs are traceable to verifiable carbon standard sources
Faithfulness	Factual adherence of generated content to retrieved source documents (Eq. (12)	R2: Technical Robustness & Safety	Directly measures hallucination prevention; high faithfulness scores demonstrate safe operation in a high-stakes regulatory reporting context
Relevancy	Topic alignment with carbon reporting domain requirements (Eq. (13)	R4: Transparency; R6: Environmental & Societal Well-being	Confirms outputs align with regulatory disclosure requirements, directly supporting climate governance objectives
Experiment Quality	Semantic similarity across 10 repeated runs; std dev analysis (Eq. (14)	R2: Technical Robustness & Safety; R7: Accountability	Run-to-run stability confirms system robustness; ablation variant confirms traceable contribution of each agent component to output quality

$$\mathrm{ROUGE}_1 = \frac{\sum_{S \in \{\mathrm{RefCRs}\}} \sum_{\mathrm{GenCR} \in S} \mathrm{Count}_{\mathrm{match}}(\mathrm{GenG}_1)}{\sum_{S \in \{\mathrm{RefCRs}\}} \sum_{\mathrm{GenCR} \in S} \mathrm{Count}(\mathrm{GenG}_1)} \tag{7}$$

- **ROUGE$_2$** (bigram overlap):

$$\mathrm{ROUGE}_2 = \frac{\sum_{S \in \{\mathrm{RefCRs}\}} \sum_{\mathrm{GenCR} \in S} \mathrm{Count}_{\mathrm{match}}(\mathrm{GenG}_2)}{\sum_{S \in \{\mathrm{RefCRs}\}} \sum_{\mathrm{GenCR} \in S} \mathrm{Count}(\mathrm{GenG}_2)} \tag{8}$$

- **ROUGE$_{lcs}$** (sentence-level longest common subsequence while ignoring the newlines)

$$\mathrm{ROUGE}_{\mathrm{lcs}} = \frac{\mathrm{LCS}(\mathrm{RefCRs},\ \mathrm{GenCRs})}{l_{\mathrm{RefCR}}} \tag{9}$$

Next, we measured **Report Quality** as the mean of indicators for **Context Recall** and **Context Precision,** two metrics recommended by Yu and colleagues in their seminal paper on retrieval-augmented generation. Our metric **Context Recall @ K**, shown in **Equation (10),** measures out of all records relevant to the report in our databases, how many relevant records were retrieved. We use this measure to find how many of the records that are present in our database have been retrieved during the Multi-Agentic Retrieval Stage correctly for generating the portion of the report.

$$\mathrm{Recall} = \frac{1}{k} \sum_{i=0}^{k} \frac{\mathrm{Rel.\ Contexts\ Retrieved}}{\mathrm{Total\ Number\ of\ Rel.\ Contexts}} \tag{10}$$

In contrast, **Context Precision @ K** evaluates the share of relevant records retrieved, out of all database records retrieved, as shown in **Equation (11)**. Unlike in Context Recall, here we focus on measuring the *correctness:* how many of the retrieved records were actually relevant. This helps us understand hallucinations being handled during the critique generation and prior context updating during the process.

$$\mathrm{Precision} = \frac{1}{k} \sum_{i=0}^{k} \frac{\mathrm{Rel.\ Contexts\ Retrieved}}{\mathrm{Total\ Number\ of\ Retrieved\ Contexts}} \tag{11}$$

Next, we measured **Faithfulness,** using a straightforward metric we designed, as shown in **Equation (12).** To avoid hallucinations in our generated texts across each section, we measure the faithfulness of our MARAG System to the information presented in the retrieved passages. So, to measure this factual consistency of the generated context to be relevant to the proprietary and public contexts, we design this metric to output a score in the range of 0 to 1, where the "1″ score means that all claims across each individual section can be found in the context either directly or through indirect interpretation while the "0" score means a completely hallucinated output.

$$\mathrm{Faithfulness} = \frac{\mathrm{No.\ of\ carbon\ claims\ in\ section\ found\ in\ context}}{\mathrm{No.\ of\ carbon\ claims\ in\ generated\ section}} \tag{12}$$

After benchmarking the contents of retrieved contexts, we measure **Relevancy,** describing how relevant a particular generated section text is in comparison to the ground truth contexts present across all our data, as shown in **Equation (13)**. We assign low scores to those generated texts that may have redundant information with the previously generated context across different sections, as well as to texts that are incomplete in nature, i.e. that have missing context. In contrast, high scores are assigned to those section texts that have concise and factual grounded information present in them. Cosine similarity, a widely used measure since Singhal (2001) quantified how well the generated text aligns with its retrieved context in terms of salient information and completeness. In this study, we use this metric as relevancy, computed at the section level to measure how well the generated text aligns with the retrieved context in terms of key information and completeness. Specifically, we embed the generated text (G) and the reference context (C) and calculate their cosine similarity:

$$\mathrm{Relevancy} = \frac{G \bullet C}{\|G\| \bullet \|C\|} \tag{13}$$

where

- G = vector representation of the generated text
- C = vector representation of the reference context
- $\|G\|$ and $\|C\|$ are the Euclidean norms of the vectors
- $G \bullet C$ represents the dot product of the two vectors

Finally, we measure the overall **Experiment Quality** of each generated AI-generated section using **semantic similarity**, a metric used in text analysis to compare how close in *meaning* two text samples are, even if they use different words, as shown in **Equation (14)**. We employ this metric to capture and compare the meaning of the generated content with ground-truth relevant test data. By performing this check, we essentially evaluate how close our generated texts match the intended meaning of the ground truth texts. Semantic similarity quantifies how closely the meaning of the generated text matches the reference text. We compute it using Zhang et al.'s BERTScore (Zhang et al., 2019) or word embedding similarity, a popular measure of semantic similarity cited over 6000 times to date:

$$\text{Semantic Similarity} = \frac{1}{N}\sum_{i=1}^{N} \max_j \text{sim}(w_i^G, w_j^C) \quad (14)$$

where:

- N = number of words in the generated text
- w_i^G = embedding of the *i*th word in the generated text
- w_j^C = embedding of the *j*th word in the reference text
- $\text{sim}(w_i^G, w_j^C)$ = cosine similarity between word embeddings

4.3. Experimental results

To assess the effectiveness and robustness of SMARAG in real-world carbon reporting tasks, we set up two experiments. The first experiment aims to validate the effectiveness of SMARAG collaborating with different LLMs, including GPT-4o, Claude-Sonnet-3.5, Gemini-1.5-Flash, DeepSeek-R1, and Mistral-Instruct-v0.3. It helps to understand how underlying model capabilities influence reporting generation performance. After identifying the most optimal LLM, the second experiment aims to benchmark SMARAG's performance against three comparative approaches, including Multi-agentic RAG (MARAG), Vanilla RAG (Single-Agent RAG), and standalone LLMs (No RAG), as shown in Table 4. In both experiments, each approach is used to generate standardized carbon reports from 2017 to 2020 for a higher education institution. Moreover, the experiments for each annual carbon report were repeated 10 times to examine the robustness of the results. The mean and standard deviation from the repeated experiments are presented in the resulting figures.

To evaluate the effectiveness of SMARAG in generating carbon reports, Fig. 7 presents the first experiment's results with the mean and standard deviation of the performance measures of SMARAG across different commercial LLMs, including GPT-4o, DeepSeek-R1, Claude-Sonnet-3.5, Gemini-1.5-Flash, and Mistral-Instruct-v0.3.

Results in Fig. 7 indicate that SMARAG, supported by GPT-4o, achieves the highest overall performance (average score of 8.1 / 10), outperforming other models in all key aspects. Smaller LLMs like DeepSeek-R1 and Mistral-Instruct-v0.3 exhibit *lower* average carbon reporting quality (5.7 vs. 6.1), suggesting that the generation of more comprehensive and complex carbon reports can benefit from *larger* reasoning models. Additionally, a strong correlation between Experiment Quality & Report Quality (Pearson's r = +82%), Usefulness & Faithfulness (r = +87%) and Usefulness & Relevancy (r = +92%) is observed across all the experiments performed, implying that more useful reports tend to be more dependent on reference information (e.g., domain and entity-specific information). Gemini-1.5-Flash demonstrates high carbon reporting quality (8.5), indicating its strong reasoning and agentic communication capabilities, though it slightly lags in other performance measures (7.2–8.1). Given GPT-4o's superior performance, it emerges as the most reliable and effective model with carbon report generation for the proposed SMARAG.

To further validate the robustness of SMARAG, the second experiment compares (1) SMARAG's overall performance with other baseline approaches, including (2) MARAG, (3) Vanilla RAG, and (4) standalone LLM. Since we have already identified that GPT-4o supports the overall optimal performance with SMARAG, this robustness experiment uses GPT-4o in SMARAG and other baseline approaches. Fig. 8 compares the resulting performance measures (Experiment Quality, Report Quality, Usefulness, Faithfulness, and Relevancy) across SMARAG, MARAG, Vanilla RAG, and standalone LLM in generating annual carbon reports for the selected higher-ed institution from 2017 to 2020. Each data point represents the average performance value from the 10 repeated experimental runs, with the +/- one standard deviation bounds.

Table 4
The systematical comparison among SMARAG and other AI-enabled carbon reporting approaches.

Modeling	RAG	Multi-Agent	Self-Decisive
LLM alone	×	×	×
Vanilla RAG	✓	×	×
MARAG	✓	✓	×
SMARAG	✓	✓	✓

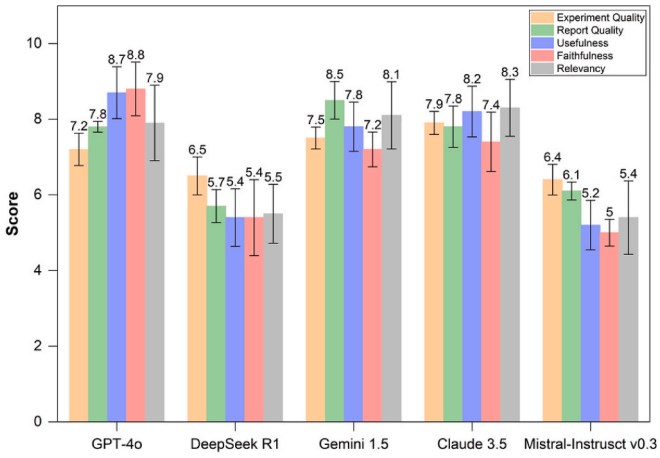


Fig. 7. SMARAG performances in carbon reporting, with different LLMs.

The results in Fig. 8 demonstrate that SMARAG consistently outperforms MARAG, Vanilla RAG and standalone LLM in generating annual carbon reports across all tested years, achieving higher scores (>6) across all evaluation metrics. For example, faithfulness is an important factor to measure the ability to prevent hallucination. A higher faithfulness score means lower hallucinations. For the 2020 annual carbon report, the faithfulness scores of SMARAG, MARAG, vanilla RAG, and standalone LLM are 8.38, 5.21, 4.12, 2.34, respectively. The faithfulness measure of SMARAG is 60.8% higher than MARAG, 103.4% higher than vanilla RAG, and 258.1% higher than standalone LLM (ChatGPT-4o).

Moreover, even the worst performances of SMARAG in the five metrics are much better than the best performances of other baseline approaches. This indicates that SMARAG can help carbon entities in their carbon reporting with significantly better quality in terms of accuracy, faithfulness, and robustness. SMARAG exhibits reliable stability for consistent performance in annual carbon reporting across the tested years (with std = 0.44), whereas MARAG (std = 0.49), Vanilla RAG (std = 0.50) and standalone LLM (std = 0.55) display marked variability and lower overall quality in their performances.

4.4. Discussions and implications

This section provides a holistic discussion of SMARAG across three dimensions: system architecture design, implementation, and experimental evaluation. Rather than reviewing individual results in isolation, the discussion examines how SMARAG's design choices collectively operationalize the EU HLEG (2019) Trustworthy AI principles within the Knowledge-Intensive Reporting Domain, situates SMARAG against alternative AI-enabled approaches, acknowledges current limitations, and draws out the broader industrial implications of Trustworthy AI-powered carbon reporting.

4.4.1. System suitability for trustworthy AI

SMARAG is designed from the ground up to operationalize six of the seven EU HLEG (2019) TAI requirements within the specific demands of the KIRD domain. Rather than treating trustworthiness as a post-hoc assurance layer, each architectural decision in SMARAG is motivated by at least one KIRD domain property and addresses at least one TAI requirement. The six requirements are addressed through a coherent

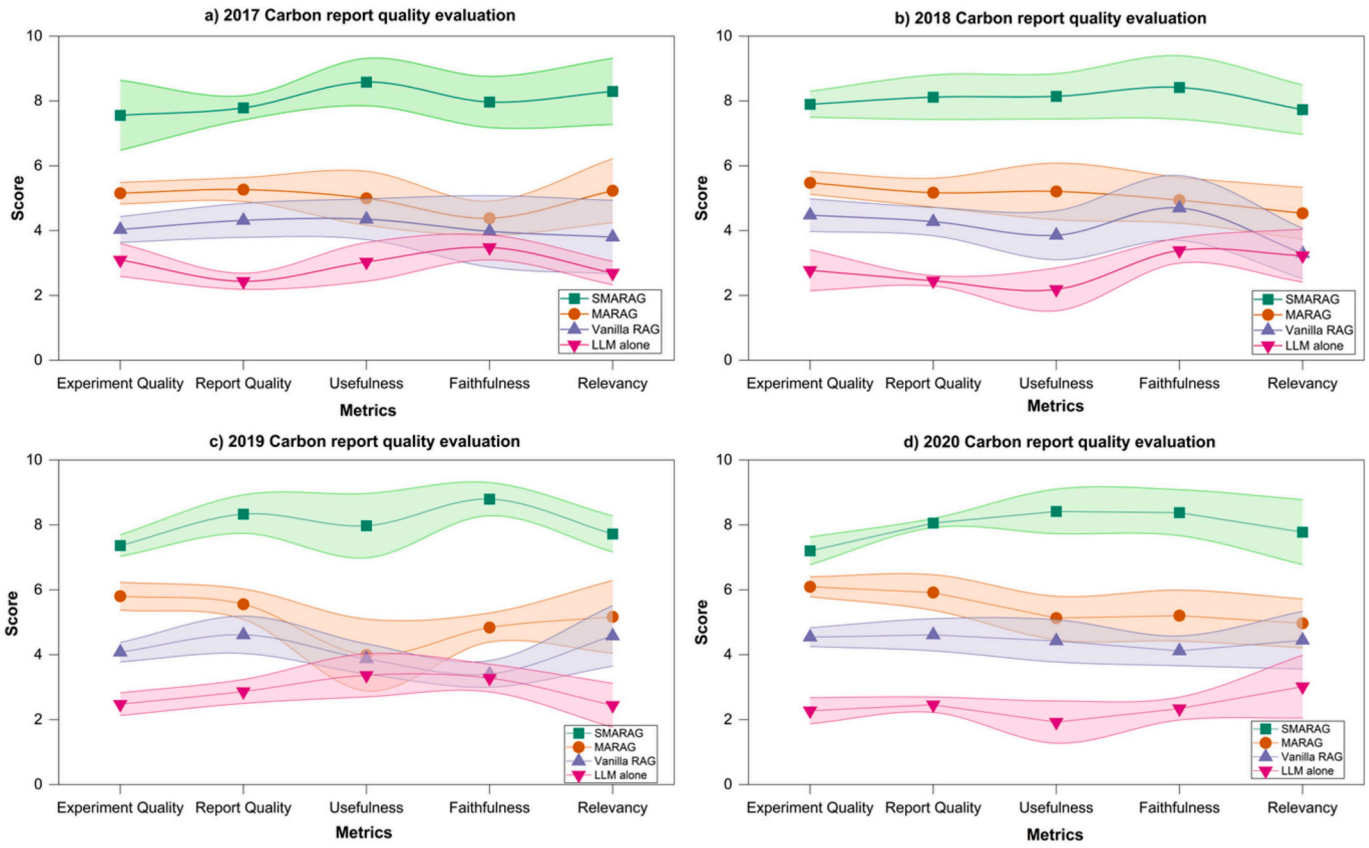


Fig. 8. Performance comparison of report generation among proposed SMARAG, MARAG, Vanilla RAG and standalone LLM approaches.

combination of architectural design and empirical evidence, establishing SMARAG as a purpose-built trustworthy AI system rather than a general-purpose language tool applied to a compliance context. Table 5 formalizes this mapping across requirements, domain properties, design responses, and empirical evidence.

Four of the six addressed requirements are grounded in quantitative experimental evidence. First, the high faithfulness scores (SMARAG 8.38 vs. standalone LLM 2.34) directly substantiate Technical Robustness and Reliability (R2). RAG reduces hallucination risk by anchoring generation in domain-specific knowledge, while the self-decisive critique loop enforces regulatory alignment before any output is finalized. Second, the same critique loop operationalizes Human Agency and Oversight (R1) by structurally embedding a regulatory review step that cannot be bypassed. Third, Accountability and Auditability (R7) is supported through the traceable agentic workflow, in which each agent's task, retrieved context, and output are logged and inspectable. Fourth,

Table 5
Trustworthy AI requirements, KIRD properties, and SMARAG design evidence.

EU HLEG Requirement	KIRD Domain Property	SMARAG Design Response	Empirical Evidence	Status
Human agency and oversight	P5-VAR; P2-MSWS	Human-in-the-loop verification gates; generation plan review before execution; iterative report appraisal	Usefulness metrics (Eq. (7)–(9): SMARAG outperforms all baselines confirming AI-generated drafts are actionable for human review (Section 4.3)	Addressed
Technical robustness and safety	P5-VAR	Self-decisive critique loops; inter-agent validation; 10-run repeated experiments for stability analysis	Faithfulness: SMARAG 8.38 vs. standalone LLM 2.34 (Eq. (12); Experiment Quality across 10 repeated runs (Eq. (14)	Addressed
Privacy and data governance	P3-PDS	Three-tier local knowledge architecture; data minimization for LLM API calls; Docker network isolation; Institutional internal cloud	Section 3.2 data security architecture; no raw proprietary data transmitted to LLM APIs	Addressed
Transparency	P1-RCD; P2-MSWS	EPA-grounded stakeholder workflow; generation plan displayed before execution; audit trail and provenance tracking; standard-aligned retrieval	Report Quality (context recall/precision, Eq. (10)–(11): SMARAG outperforms all baselines confirming standard-aligned retrieval; Relevancy metric confirms domain alignment (Section 4.3, Table 3)	Addressed
Diversity, non-discrimination and fairness	N/A	Not addressed in current study	N/A	Future research
Environmental and societal well-being	P1-RCD; P4-TEKD	SMARAG directly enables accurate climate disclosure aligned with Paris Agreement and national net-zero commitments; reducing cost barriers to carbon reporting participation	Section 4.3 Discussions; reduced reporting cycle from months to minutes; broader resource-constrained organizations' participation	Addressed
Accountability	P2-MSWS; P5-VAR	Role-specific agents with defined responsibilities; experiment quality metric; human oversight before report finalization	Experiment Quality (Eq. (14): ablation confirms traceable agent contributions (Section 4.3); Usefulness metric confirms human-auditable output quality (Table 3)	Addressed

Transparency and Explainability (R4) is addressed through the structured service layer, which presents outputs in stakeholder-appropriate formats linked to retrieved regulatory evidence, making the basis for each disclosure claim reviewable.

Two further requirements are substantiated through architectural design rather than quantitative metrics. Specifically, Privacy and Data Governance (R3) is operationalized through data locality. The proprietary carbon data is stored exclusively on organization-local servers, and no sensitive information is transmitted to external LLM API endpoints. Network isolation is enforced through HTTPS/TLS and SSL termination at the infrastructure layer, as documented in Section 3.4. Finally, Societal and Environmental Well-being (R6) is addressed through SMARAG's explicit carbon reporting mission and its alignment with EPA guidelines and international climate disclosure frameworks, ensuring that outputs support the environmental governance objectives that carbon reporting is designed to serve.

The requirement of Diversity, Non-Discrimination, and Fairness (R5) remains unaddressed in the current system design. SMARAG does not yet incorporate mechanisms to audit for representational bias in its knowledge base or to ensure equitable service quality across organizations of different geographies, languages, or economic contexts. This gap is acknowledged transparently in Table 5 and discussed further in Section 4.4.3.

4.4.2. Comparison with other AI-Enabled carbon reporting approaches

To frame the TAI suitability of SMARAG, we can consider how progressively more capable AI-enabled approaches, either addressed or further failed to address, the trustworthiness needs of the KIRD. The four approaches introduced in section 4.3, represent a natural hierarchy of increasing sophistication in architectural design: standalone LLM, Vanilla RAG, MARAG, and SMARAG.

A standalone LLM for carbon reporting generates text from parametric knowledge alone, with no retrieval grounding or domain-specific regulatory alignment. We obtain the lowest faithfulness scores from the experimental evaluation with this variant (2.34), also highlighting the commonly known trend that LLMs hallucinate in knowledge-intensive domains (Durmus et al., 2020). From a TAI perspective, standalone LLM approach fails to address R2 (no retrieval grounding), R1 (no structured oversight), R3 (proprietary inputs may be exposed through API calls), and R7 (generation is non-traceable). Given that we are in the KIRD where users are accountable to regulators and the public for what outcomes are produced, the standalone LLM approach cannot be trustworthy.

Vanilla RAG, a single agent RAG, improves on the standard failure by introducing grounding in retrieval of domain specific documents, causing a marked improvement in the faithfulness and relevance scores against the standalone LLM baseline. However, Vanilla RAG lacks the self-decisive critique loop that evaluates generated content against regulatory standards before finalization, meaning that retrieved context may be applied inconsistently or incompletely without detection. Standard RAG configurations also typically transmit proprietary query content to external API endpoints, leaving R3 unaddressed. Vanilla RAG somewhat advances on R2 with the grounding of knowledge by documents retrieved but remains weak in terms of architectural properties at R1, R3, and R7.

MARAG builds on top of Vanilla RAG by expanding its architecture to contain a “multi-agent system”, capable of performing the carbon reporting task via a number of specialized agents responsible for independently writing a discrete section of the carbon report. This improves task decomposition and report quality, as reflected in the performance gains over Vanilla RAG in Fig. 8. However, MARAG does not incorporate a self-decisive critique mechanism that checks outputs against regulatory standards before acceptance, nor does it implement data locality controls or a structured human oversight architecture. MARAG therefore advances R2 further but does not substantively address R1, R3, R4, or R7 as architectural properties.

SMARAG extends the MARAG foundation with three targeted additions that close the remaining TAI gaps: the self-decisive critique loop (R1 and R7), data locality and encrypted transmission (R3), and a structured service layer that maps outputs to stakeholder roles and regulatory sources (R4). The experimental results confirm that these additions do not compromise performance. SMARAG achieves the highest scores across all five metrics while simultaneously satisfying the broadest set of TAI requirements. This progression demonstrates that trustworthiness and performance are complementary rather than competing design objectives in the knowledge-intensive domains.

4.4.3. Current limitations and adoption barriers

Despite the merits, several limitations of the current study should be acknowledged. First, SMARAG's output quality depends on the completeness and internal consistency of proprietary data supplied by reporting entities. Incomplete and/or missing records of emissions may lead to an incomplete emissions report, which will represent a gap in the generated critique through the critique function but are not resolved without the intervention of a human user. This reflects the dependency on the quality of the data in any retrieval augmentation approach. This limitation is characteristic of the KIRD more broadly, where data quality varies substantially across organizations and disclosure cycles.

Second, experimental validation is conducted within a single-domain context, higher education institutional carbon disclosure based on Second Nature data. Even though this information is sufficient to prove the effectiveness of SMARAG, the findings may not generalize to other domains that have far more diverse sources of emissions and more complicated operating processes. The SMARAG architecture, based on KIRD properties and the TAI principles, is intended to be domain-transferable, but we still need empirical validation across further disclosure contexts.

Third, requirement R5 (Diversity, Non-Discrimination, and Fairness) of the EU HLEG is not operational within SMARAG. Currently, there is no auditing representational bias within our knowledge base or seeking equitable service quality in carbon reports. Closing this gap, by conducting fairness audits on the developing knowledge base, supporting multiple languages and including multiple stakeholders in the development of the solution have been listed as a key area for future development for other sustainability reports to reach full TAI compliance.

Beyond the technical limitations identified above, the broader adoption of AI-powered carbon reporting systems faces structural barriers across three dimensions: socio-economic barriers, technical barriers, and organizational and workflow inertia (Boonmee, Mangkalakeeree and Jeong, 2025; Moghayedi, Michell and Awuzie, 2024). From a socio-economic perspective, adoption is heavily conditioned by the regulatory environment in which organizations operate. In jurisdictions where carbon disclosure requirements remain limited or loosely enforced, the financial justification for investing in AI-assisted reporting infrastructure is weak for many organizations, particularly resource-constrained small and medium-sized enterprises (SMEs), given the associated costs in human resources, digital infrastructure, and professional services. (Moghayedi, Michell and Awuzie, 2024) further highlight policy inconsistencies and misaligned stakeholder incentives as major barriers to AI adoption in climate-related applications, underscoring that technological readiness alone is insufficient where regulatory incentive structures are absent or fragmented. From a technical perspective, three technical concerns also matter. First, carbon data in the real world is in non-standardized formats, inconsistent scope boundaries, and heterogeneous sources, making integration into AI pipelines difficult, let alone alignment with mixed and evolving frameworks such as CDP, GHG Protocol, and ISO standards (Mustafa et al., 2025). Second, concerns around LLM hallucination and factual unreliability in high-stakes regulatory contexts have additionally impeded organizational trust in AI-generated compliance outputs (Yao et al., 2024). Third, sensitivities around proprietary emissions data introduce further hesitancy toward externally hosted AI deployments. From an

organizational perspective, carbon reporting has historically been conducted by domain experts and specialist consultants over extended timelines. The introduction of agentic AI systems requires not only workflow redesign but also institutional trust in automated outputs, reskilling of personnel, and alignment across multi-stakeholder reporting chains. Organizational alignment and change management capacity, rather than technical capability alone, are increasingly recognized as the binding constraints on AI adoption for sustainability (Schwaeke et al., 2025). Together, these three categories of barriers suggest that the diffusion of trustworthy AI in carbon reporting will depend not only on system performance but on parallel advances in regulatory standardization, digital infrastructure accessibility, and organizational change management.

4.4.4. Industrial implications

SMARAG represents a significant AI booster for carbon reporting practice, addressing the three core challenges identified in Section 1. The experimental results confirm that SMARAG substantially mitigates hallucination through RAG-grounded generation and the self-decisive critique loop, achieving faithfulness scores (8.38) far exceeding standalone LLM baselines (2.34). The multi-agentic architecture decomposes the complex carbon reporting workflow into coordinated specialist agents, enabling reliable handling of multi-stakeholder obligations without loss of traceability. Critically, SMARAG achieves these gains while keeping proprietary emissions data local, ensuring regulation-compliant operation without compromising enterprise data security.

Built upon its technical rigor, SMARAG holds substantial potential to catalyze inclusive participation in climate action from various sectors of different sizes and levels of economic entities that all have a role to play in fighting climate change. For any carbon reporting entity, SMARAG significantly reduces both the financial burden and technical barriers to high-quality carbon disclosure. In current practice, sustainability report generation is managed by consulting firms and technically trained sustainability experts, which takes months to complete an acceptable report (Aiuto, Huckins and Momblanco, 2024; Liu et al., 2021; Wu et al., 2022). SMARAG compresses this process to minutes, making meaningful participation in international carbon markets and disclosure regimes feasible for resource-constrained organizations without proportionally larger compliance budgets or specialized staff. For instance, SMARAG holds substantial potential to support small and medium-sized enterprises (SMEs) in fulfilling their carbon disclosure obligations. While the current validation is conducted on higher education institutions (HEIs), which serve as a structurally analogous proxy in that they similarly lack dedicated sustainability reporting teams and face comparable resource constraints. Its stakeholder-aware agentic workflows, configurable data integration pipeline, and privacy-preserving RAG infrastructure can, in principle, be directly adapted to any carbon-reporting entity regardless of sector or size. Empirical validation with SME-specific datasets constitutes an important direction for future research. Subject to such validation, SMARAG could enable wider participation from more entities in a more inclusive and credible climate governance.

For the carbon market, SMARAG directly contributes to a more transparent, trustworthy, and investable carbon economy. By producing standardized, regulation-aligned, and hallucination-resistant carbon reports, SMARAG improves the verifiability and comparability of disclosed emissions data, strengthening investor confidence and directing sustainable finance toward credible projects. Moreover, SMARAG's ability to reduce reporting costs and automate stakeholder workflows holds promise for encouraging broader corporate participation in climate action, particularly among SMEs that have historically faced financial and operational barriers to climate disclosure, a generalization that warrants dedicated empirical investigation in future work.

At the societal level, SMARAG enhances the infrastructure of climate governance by enabling the generation of accurate, verifiable, and policy-aligned carbon disclosures. The system's agentic architecture, grounded in EPA guidelines (EPA, 2024) and designed under EU HLEG TAI principles, bridges the gap between regulatory ambition and practical implementation. By embedding human oversight at each tier of the reporting pipeline and linking structured review to both plan validation and output appraisal, SMARAG supports the institutionally credible and auditable disclosure process that climate policymakers and international frameworks increasingly require. Its deployment across sectors and organizational scales may therefore contribute to the broader systemic goals of climate transparency, market credibility, and governance accountability.

5. Conclusion and future research

This paper introduces SMARAG, a trustworthy AI system for carbon reporting, using Self-decisive Multi-Agentic Retrieval-Augmented Generation. The theoretical foundations are established by mapping six of the seven EU HLEG TAI requirements and five Knowledge-Intensive Reporting Domain (KIRD) properties (P1–P5) into SMARAG's design. Building on the TAI requirements and KIRD properties, a seven-layer system architecture was designed, including three-tier human oversight, standards versioning, and privacy-by-design data management, throughout the full reporting stack, from the supply chain scenario layer to the stakeholder layer. The system was implemented with a Self-Decisive Task Decomposition Engine and a Multi-Agentic Carbon Reporting Workflow, integrating five LLMs (GPT-4o, DeepSeek-R1, Claude-Sonnet-3.5, Gemini-1.5-Flash, and Mistral-Instruct-v0.3) within a secure, locally deployed infrastructure. Evaluated against MARAG, vanilla RAG, and LLM-only baselines on real-world carbon data from Second Nature–affiliated institutions, SMARAG achieved consistent superiority across all five TAI-grounded evaluation metrics, such as faithfulness, report quality, usefulness, experiment quality, and relevancy, while compressing reporting timelines from months to minutes.

This study makes four key contributions to the literature on AI-assisted climate governance. **First**, it provides one of the first Trustworthy AI systems in carbon reporting, following the EU HLEG Ethics Guidelines. Five domain properties in KIRD are introduced to formalize the unique requirements of carbon reporting as a knowledge-intensive domain and provide a principled, theoretically grounded bridge between TAI design principles and carbon disclosure practice. **Second**, it advances multi-agentic RAG approach through a self-decisive task decomposition mechanism in which agents dynamically plan, decompose, and critique report generation tasks at runtime, moving beyond fixed pipelines toward domain-specific, regulatory-aware agentic design. The system maps real-world stakeholder roles to autonomous agents with role-specific evaluation criteria and embeds a three-tier human oversight structure (Human-before-the-Loop, Human-in-the-Loop, Human-on-the-Loop) that operationalizes human agency throughout the automated reporting workflow. **Third**, SMARAG establishes privacy-by-design architecture for AI-assisted carbon reporting, combining local knowledge repository deployment, data minimization for LLM API calls, and structured audit trail generation. It addresses the data security and confidentiality concerns that have impeded broader AI adoption in corporate sustainability reporting. **Fourth**, the system advances inclusive climate governance by demonstrating that high-quality, regulation-aligned carbon disclosure is achievable without the months-long engagement of specialized consultants, enabling resource-constrained organizations to participate meaningfully in credible climate governance.

Future research can extend SMARAG in four dimensions. From a domain dimension, the TAI-grounded architecture can be adapted for broader sustainability reporting, including ESG disclosures, corporate social responsibility reporting, enabling automated, standardized, and context-aware disclosure across diverse regulatory frameworks. From a lifecycle dimension, AI-empowered carbon auditing can be explored to automatically verify the emissions information provided by reporting entities, improving the credibility of disclosures and supporting the formation of robust, verifiable carbon markets. From an organizational

dimension, future research should empirically validate SMARAG's performance on SME-specific carbon datasets; while the current study uses higher education institutions as a structurally analogous proxy for resource-constrained organizations, SME-focused validation would provide direct evidence of generalizability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data and the analytical codes can be accessed through the GitHub link: https://github.com/Levis62/SMARAG-Data-Sources.

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